

BMEG3920 • ONE-PAGE PITCH

LiteRehab Fusion

Immediate feedback for upper-limb rehabilitation practice at home.

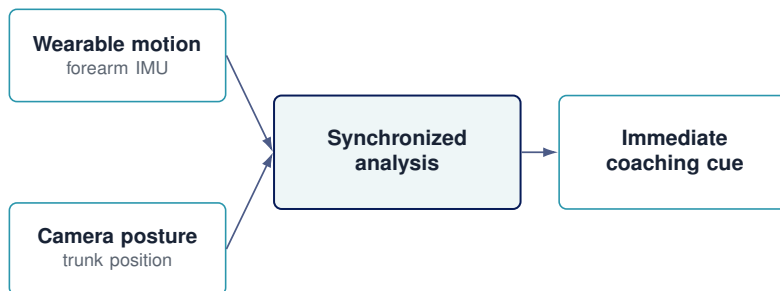
Feedback arrives too late

A patient may leave a physiotherapy appointment knowing what to practise. At home, it is harder to tell whether the movement was controlled and large enough, or whether leaning did part of the work. By the next appointment, those repetitions are already behind them.

LiteRehab responds during the repetition

LiteRehab gives a cue while the exercise is still happening. A forearm wearable tracks motion, while a camera checks posture. Together, they identify the demonstrated exercise, count repetitions, and flag excessive speed, limited range, or trunk compensation. The dashboard also keeps a synchronized session record.

Motion and posture, seen together



"Slow down" ■ "Increase the range" ■ "Keep the trunk still"

Forearm wearable (MPU6050 + ESP32) ■ BLE receiver ■ MaixCAM2 posture stream ■ computer dashboard

The wearable sees movement. The camera sees posture.

WEARABLE MOTION

Repetition timing and movement range from the forearm.

CAMERA POSTURE

A wider view that can flag trunk compensation.

WORKING PROTOTYPE

What the prototype does today

- Recognises the demonstrated elbow flexion and forearm rotation exercises.
- Gives feedback for excessive speed, insufficient range, and trunk compensation.
- Shows the repetition count and records synchronized motion and posture data.
- Uses a classroom CNN-BiGRU baseline with a rule-based fallback.

A cue now, a record for later

FOR THE PERSON PRACTISING

A short cue arrives in time to adjust the next repetition.

FOR THE PHYSIOTHERAPIST

A session record could show what happened between appointments. Exercise selection and clinical decisions remain with the physiotherapist.

Next: supervised usability testing

Our next step is to test the workflow with physiotherapists and representative users. Their feedback would help us refine the prompts and collect movement data under professional supervision.

COURSEWORK PROTOTYPE • NOT A MEDICAL DEVICE

LiteRehab Fusion does not diagnose, prescribe treatment, score recovery, or replace professional supervision. The current system supports a limited exercise set and makes no clinical accuracy claim.